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Sharps injuries and splash exposures among healthcare workers in Arab countries: protocol of a systematic review and meta-analysis

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Sharps injuries and splash exposures among healthcare workers in Arab countries: protocol of a systematic review and meta-analysis

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ABSTRACT

Introduction Sharps injuries, including needlestick injuries, and splash exposures constitute serious occupational health problems for healthcare workers, carrying the risk of blood-borne infections. However, data on such occupational incidents and their risk factors in healthcare settings are scarce and not systematically summarized in the Arab countries. The aim of this study is to conduct a systematic review and meta-analysis to review published literature about sharps injuries and splash exposures of healthcare workers in Arab countries, with the objectives to determine the incidence and/or prevalence of these events, their identified risk factors and the applied preventive and postexposure prophylactic measures.

Methods and analysis The protocol is developed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol (PRISMA-P) guidelines. A comprehensive pre-search developed in January to March 2021 in the database PubMed will be followed by a systematic search of six, core medical and health science databases; PubMed, EMBASE, Scopus, CINAHL, Web of Science and Africa-Wide Information in May 2021. The search will be performed without any filters or restrictions for publication years. Covidence systematic review tool will be used for document management, blinded screening and study selection. Two reviewers will independently screen the records, extract data and conduct risk of bias assessment. Results will be synthesized narratively in summary tables, and, if findings allow, meta-analysis will be conducted on the incidence and/or prevalence of sharps injuries and splash exposures, and on the effect size of risk factors.

Ethics and dissemination The systematic review methodology does not require ethics approval due to the nature of the study design based only on published studies. The results of the systematic review will be published in a peer-reviewed journal, disseminated to stakeholders and made publicly available.

Prospero registration number CRD42021242416

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Strengths and limitations of this study

- Being the first comprehensive systematic review on sharps injuries and splash exposures among healthcare workers in Arab countries, it will provide gap-filling data for further improvement of risk management.
- The study will rigorously follow the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
- The study will use transparent screening, data extraction and analysis methodology carried out by several researchers knowledgeable in conducting systematic reviews.
- Meta-analysis, including subgroup and sensitivity analysis, will be conducted if suitable results can allow for that.
- Scarcity of data and high clinical heterogeneity can be the main limitations of the study.

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INTRODUCTION

A sharps injury is a penetrating stab wound from a sharp object, such as a needlestick injury.[1] It is a very common event among healthcare personnel and, together with splash exposures on mucus membranes, constitute the most efficient method for transmitting blood-borne infectious agents from patients to healthcare staff. Sharps injuries are primarily associated with occupational transmission of hepatitis B (HBV), hepatitis C (HCV) and human immunodeficiency virus (HIV), but may play role in the spread of other infectious agents, such as coronaviruses, as well. Health care personnel are at increased risk of sharps injuries and splash exposures because of the high frequency of contacting blood and other body fluids and potentially contaminated sharp objects.[2] As a consequence of the disease burden of occupational blood-borne infections, such incidents can entail a number of different costs including the loss of employee time, the cost of tying up staff to investigate the injury, expense of laboratory testing, the cost of postexposure prophylaxis and treatment for exposed and infected workers, expenses of temporarily or permanently replacing staff, and so on.[1-3]

According to the World Health Report 2002 by the World Health Organization,[4] about two million healthcare workers sustain percutaneous exposure to infectious diseases every year. The highest frequency of needlestick injuries has been reported among nurses, because they are the most exposed to needles and other sharp instruments during their work.[5]

There are several studies, mostly published on sharps injuries in healthcare settings from Europe and America. A comprehensive study estimated annual number of sharps injuries to be 384,000 in the United States, 100,000 in the United Kingdom, 700,000 in Germany, 29,719 in France, 28,200 in Italy and 21,815 in Spain.[6] There are reports about the high frequency of sharps injuries from other parts of the world, too. Nearly one out of five healthcare workers had experienced needlestick or sharps injury at least once within the past one year according to the results of a cross-sectional study that was conducted in 2014 in four hospitals in Southeast Ethiopia.[7] In a large-scale hospital-based survey in China, 6.3% of healthcare workers sustained sharps injury in the last month, corresponding to 1,032 incidents per 1,000 employees yearly.[8] The majority of injuries were caused by hollow-bore needles (63%) but only 4.3% of the incidents were reported to the infection control team. Another, yet unpublished study from China also found a high (41%) 5-year prevalence of sharps injuries among hospital personnel and a substantial (over 80%) underreporting was revealed (unpublished data).

However, limited information exists about sharps injuries and splash exposures of healthcare workers in the Middle East and North Africa. Several studies were published on this topic from Iran and their results were summarized in a systematic review.[9] However, the information on sharps injuries and splash exposures in Arab countries is limited and it has never been systematically reviewed. The results of an anonymized

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questionnaire survey among healthcare workers in the United Arab Emirates showed that 19% of the respondents had sustained a sharps injury in the calendar year 2006 and 53% of these sharps injuries were caused by blood contaminated sharp objects.[10] 35.6% of Egyptian healthcare workers were reported in 2003 to be exposed to needlestick injury during the past three months, with an estimated annual number of nearly five needlesticks per worker.[11]

The Arab World consists of 22 countries in the Middle East and North Africa: Algeria, Bahrain, the Comoros Islands, Djibouti, Egypt, Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, Mauritania, Oman, Palestine, Qatar, Saudi Arabia, Somalia, Sudan, Syria, Tunisia, the United Arab Emirates, and Yemen. Although Arab countries constitute a heterogeneous group by economic and social development and consequently by the structure and development of their healthcare systems, it is a general phenomenon that healthcare employees work in fast-paced, stressful and frequently understaffed environment, that has also been affected by the ongoing COVID-19 pandemic. They have to perform various tasks, use sharp objects and can get easily into contact with body fluids. Their risk for sustaining sharps injury or splash exposure is high and probably severely underreported. To improve the occupational health and safety management system of this important workplace hazard, the systematic review and synthesis of the limited existing information from the region is indispensable.

OBJECTIVES

The aim of the study is to perform a systematic review by collecting and synthetizing the information available in the scientific literature about occupational sharps injuries, including needlestick injuries, and splash exposures occurring to healthcare workers engaged in clinical patient care in Arab countries, and to conduct meta-analysis if findings make it possible.

To achieve the aim, the following objectives will be pursued:

1. To conduct a systematic review of in extenso research articles published in peer-reviewed scientific journals in English or Arabic on occupational sharps injuries and splash exposures among healthcare workers engaged in clinical patient care in Arab countries.
2. To summarize incidence and/or prevalence of occupational sharps injuries and splash exposures among healthcare workers engaged in clinical patient care in Arab countries.
3. To identify risk factors statistically significantly associated with the occurrence of occupational sharps injuries and splash exposures.
4. To summarize the existing preventive and postexposure prophylactic measures used in the management of sharps injuries and splash exposures in healthcare settings in Arab countries.

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If findings allow for conducting meta-analysis, the following objectives will also be fulfilled:

5. To compute pooled frequency estimates for the incidence and/or prevalence of occupational sharps injuries and splash exposures among healthcare workers engaged in clinical patient care in Arab countries, with the possibility of subgroup analysis.
6. To compute pooled effect size estimates for potential risk factors of occupational sharps injuries and splash exposures, with the possibility of subgroup analysis.

METHODS AND ANALYSIS

This protocol follows the Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocol (PRISMA-P) guidelines.[12] The completed PRISMA-P check list is available in Supplementary file 1. The protocol is registered online on the international prospective register of systematic reviews, PROSPERO: CRD42021242416. The final review will be informed by Cochrane Handbook for Systematic Reviews of Interventions and follow the new PRISMA 2020 statement.[13 14]

Eligibility criteria

A PECO (population, exposure, comparator and outcomes) statement is developed to review information on occupational sharps injuries and splash exposures occurring to healthcare workers engaged in clinical patient care in Arab countries.

Population

Inclusion: All workers employed in any full- or part-time jobs, including students training, performing clinical care for patients, sterilization or cleaning activities in in- or outpatient healthcare settings (referred to as healthcare workers) located in Arab countries.

Exclusion: Workers not performing clinical care, sterilization or cleaning activities in in- or outpatient healthcare settings, and working in other than Arab countries.

Exposure

Inclusion: Workplace accidents caused by sharp objects, including needles, or by splash of liquid potentially contaminated with body fluids, occurring in healthcare settings, and the risk factors of such accidents.

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Exclusion: Accidents occurring not in healthcare settings, not caused by sharp objects or by splash of liquid potentially contaminated with body fluids.

Comparator

There will be no comparator for the main outcome of the review as it is the incidence/prevalence of occupational sharps injuries and splash exposures among healthcare workers in Arab countries. Also, there will be no comparator for the additional outcome of preventive and postexposure prophylactic measures used in the management of sharps injuries and splash exposures in healthcare settings in Arab countries.

The comparator for the additional outcome of risk factors of sharp injuries and splash exposures will be healthcare workers who have not sustained sharps injury or splash exposure.

Outcome

Injuries in the form of penetrating stab wounds on any sites of the body, and splash exposures, sustained by healthcare workers during an occupational accident caused by sharp objects, primarily needles, used in the work process or being present in the workplace, or by splash of liquids potentially contaminated with body fluids during the work process. The main outcome of the review will be the incidence and/or prevalence of occupational sharps, including needlestick, injuries and splash exposures among healthcare workers in Arab countries. Additional outcomes will be risk factors identified by statistical analysis to be significantly correlated to the occurrence of occupational sharps injuries and splash exposures in healthcare settings, and their effect size; and preventive and postexposure prophylactic measures used in the management of sharps injuries and splash exposures in healthcare settings in Arab countries.

Types of studies

Observational occupational epidemiological studies will be used that apply collection of individual data to determine the incidence and/or prevalence (cohort and cross-sectional studies) and identify risk factors (cohort and case-control studies) of occupational sharps injuries and splash exposures among healthcare workers in Arab countries. Other descriptive observational study designs (ecological, case series, case report) will also be included to review the preventive and postexposure prophylactic measures.

Information sources

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PubMed (NML), Scopus (Elsevier), Web of Science (Core Collection), EMBASE (Elsevier), CHNAHL (EBSCO) and Africa-Wide Information (EBSCO) electronic databases will be systematically searched to identify eligible studies published in peer-reviewed scientific journals.

Search strategy

A comprehensive, systematic search for literature will be conducted in May 2021 by a medical librarian specialized in systematic reviews (LÖ). Preliminary search term combinations have been identified and tested based on the PECO statement and inclusion and exclusion criteria outlined above. Pre-searches in PubMed was performed in January-March 2021 (LÖ). PubMed's MeSH was used to support a systematic inclusion of the search terms and their variations. Subject experts (BA, IK & IE) reviewed the final search term selection. The search string developed in PubMed will later be translated from English to Arabic and applied in all selected databases without geographical or publication year restrictions. We will use a combination of the search fields "Title/Abstract" and MeSH (alternatively Thesaurus or Subject Headings) for the best possible information retrieval. A search field coveting 'Title", "Abstract" and "Keywords" will be used in the absence of a MeSH, Thesaurus or Subject Headings.

The pre-search in PubMed can be found in Supplementary file 2. Reproducible search strings with result and notes for all databases included in the review will be appended to the final study. De-duplication and screening details will be reported in a PRISMA flow-diagram. A complete search-update of all databases will be performed before the final analysis and hand screening of the references lists in the included studies conducted.

Study records

Data management

The citations of the search results will be imported into the systematic review software Covidence for management and blinded screening of all records identified in the literature search.[15] The modules for automatic de-duplication, title/abstract and full text screening and blinded conflict resolving will support our work.

Selection process

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The screening and selection process will be documented in a PRISMA flow-diagram.[12] After automatic duplication removal in Covidence, the unique studies retrieved will be screened in two stages by two independent reviewers (IK and IE) based on the predetermined inclusion and exclusion criteria. In the first stage, title and abstract of publications will be screened. In the second stage, the full texts of publications selected during the first stage will be considered for inclusion also by two independent reviewers (IK and IE). Reason for exclusion will be documented in a PRISMA flow diagram. Discrepancies between the judgements of the two reviewers regarding the eligibility of studies and reason for exclusion will be resolved in the software by a third reviewer (BA). The screening and conflict resolving modules in Covidence is blinded.

Data collection process

Two reviewers (IK and IE) will independently extract data using a data extraction sheet developed in Excel 2019 for this study and pilot tested before use. Discrepancies will be resolved and extracted data validated through discussions with other reviewers (BA, MSP, RHA) until convergence and agreement is reached.

Data items

At a minimum, information will be extracted on publication characteristics (title, name of the first author, year of publication, DOI), study design (study type, risk factors, period of data collection, type of measurement of exposure and outcome, statistical analysis performed and adjustment to confounders), study population (demographic and occupational characteristics), results (incidence/prevalence of sharps injuries and splash exposures, crude and adjusted effect estimates of association identified as risk factors, preventive and postexposure prophylactic measures), and conflict of interest (declaration, funding).

Risk of bias in individual studies

The risk of bias in the included studies will be assessed using a modified version of the Navigation Guide RoB tool that is tailored specifically to systematic reviews of occupational health studies.[16] The Navigation Guide RoB tool assesses risk of bias in the following domains: selection bias, ascertainment bias, accuracy of exposure and outcome assessment, confounding, incomplete data and selective reporting, conflict of interest, and other biases, which will be completed with domains from the RoB-SPEO tool that was developed for the WHO/ILO Joint Estimates of the Work-related Burden of Disease and Injury project to assess risk of bias in studies estimating the prevalence of exposure to occupational risk factors.[17] Two

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reviewers (MSP and RHA) will independently assess the risk of bias in the included studies. Discrepancies between the judgements of the two reviewers regarding risk of bias of studies will be resolved by a third reviewer (BA).

Data synthesis

First, study findings will be synthesized narratively in summary tables regarding the main (incidence and prevalence of occupational sharps injuries and splash exposures) and the additional outcomes (risk factors and preventive and postexposure prophylactic measures).

If findings allow, meta-analysis will be conducted on the incidence and prevalence of occupational sharps injuries and splash exposures, and on potential risk factors. If we find two or more studies with eligible estimates on outcome frequency and/or on risk factor effect, two reviewers will independently investigate the clinical heterogeneity of the studies. If frequency and/or effect estimates are found clinically homogenous, then studies will be combined into one pooled frequency and/or effect estimate. Statistical heterogeneity of the studies will be tested using the I^2 statistic.[18] If two or more clinically homogenous studies are found to be sufficiently statistically homogenous to be combined in a meta-analysis, the frequency and/or effect estimates will be pooled in a quantitative meta-analysis, using the inverse variance method with a random-effects model to account for cross-study heterogeneity.[18] The meta-analysis will be conducted in RevMan software.[19] The pooled estimates will be presented in forest plots.

Analysis of subgroups or subsets

If there is evidence for differences in frequency and/or effect estimates by population (country, sex, age, job, healthcare sector, or by a combination of these variables), sub-group analyses by the relevant variable or combination of variables, will be conducted. Sensitivity analyses will be performed including studies judged to be of "low" or "probably low" risk of bias and judged to be of "low" or "probably low" risk of bias from conflict of interest.

Quality of cumulative evidence

A minimum of four reviewers will assess the quality of evidence for the entire body of evidence using the Navigation Guide quality of evidence assessment tool that is based on the GRADE approach and modified to be applicable in the fields of occupational and environmental health.[20-22]

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Patient and public involvement: No patient or members of the public will be included in this study.

Ethics and dissemination: Due to the nature of this study as a systematic review, ethics approval is not necessary as no patient or members of the public will be included in the study.

The results of this review will be published in a peer-reviewed journal and will be made publicly available in electronic format. In addition, if the findings of the review warrant a change in the practice of sharps injury and splash exposure management, a summary report will be disseminated to leading healthcare and occupational health policy makers in the East Mediterranean Region of the World Health Organization, and in the United Arab Emirates.

Contributors: BA, IK, IE, MSP, RHA, MSH and LÖ were involved in conceptualization and protocol development. BA, LÖ, IK and IEB have developed the search strategy. Literature search will be conducted by LÖ and IK. Screening will be performed by IK and IEB, with BA resolving conflicts. IK and IEB will extract data, which will be validated by BA, MSP and RHA.

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Competing interests: None declared.

Word count: 2674

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Supplementary file 1

Reporting checklist for protocol of a systematic review and meta-analysis

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	N/A
Sponsor	#5b	Provide name for the review funder and / or sponsor	N/A
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	N/A
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	5
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5,6

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Methods

Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6,7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9,10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10

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Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9,10
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10

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Supplementary file 2

Pre-search in PubMed

Source: PubMed (NLM)

Search date: 2021-03-11

Search specifications: All search terms are searched in the search fields: “title” and “abstract” (here marked with TI/AB) and in MeSH (when available). No filters or limitations applied.

Result: 309 records

Search string

(("Middle East"[Mesh:NoExp] OR "Bahrain"[Mesh] OR "Iraq"[Mesh] OR "Jordan"[Mesh] OR "Kuwait"[Mesh] OR "Lebanon"[Mesh] OR "Oman"[Mesh] OR "Qatar"[Mesh] OR "Saudi Arabia"[Mesh] OR "Syria"[Mesh] OR "United Arab Emirates"[Mesh] OR "Yemen"[Mesh] OR "Arab World"[Mesh] OR "Africa, Northern"[Mesh] OR "Somalia"[Mesh] OR "Djibouti"[Mesh] OR "Comoros"[Mesh] OR "Sudan"[Mesh] OR "Mauritania"[Mesh] OR "Middle East*"[Title/Abstract] OR "Near East"[Title/Abstract] OR "East Mediterranean" [Title/Abstract] OR "Eastern Mediterranean"[Title/Abstract] OR Arabic[Title/Abstract] OR Arabs[Title/Abstract] OR Arab[Title/Abstract] OR MENA[Title/Abstract] OR "Arabian Peninsula"[Title/Abstract] OR "North Africa" [Title/Abstract] OR "Northern Africa"[Title/Abstract] OR Maghreb[Title/Abstract] OR Maghrib[Title/Abstract] OR Egypt*[Title/Abstract] OR Jordan*[Title/Abstract] OR Lebanon[Title/Abstract] OR Lebanese[Title/Abstract] OR Syria*[Title/Abstract] OR Algeri*[Title/Abstract] OR Libya*[Title/Abstract] OR Morocco*[Title/Abstract] OR Tunis*[Title/Abstract] OR UAE[Title/Abstract] OR "United Arab Emirates"[Title/Abstract] OR Saudi*[Title/Abstract] OR KSA[Title/Abstract] OR Qatar*[Title/Abstract] OR Oman*[Title/Abstract] OR Emirat*[Title/Abstract] OR Yemen*[Title/Abstract] OR Kuwait*[Title/Abstract] OR Bahrain*[Title/Abstract] OR Somail*[Title/Abstract] OR Djibouti*[Title/Abstract] OR Comoro*[Title/Abstract] OR Comorian*[Title/Abstract] OR Iraq*[Title/Abstract] OR Palestin*[Title/Abstract] OR “West Bank” [Title/Abstract] OR “Gaza Strip” [Title/Abstract] OR

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Sudan*[Title/Abstract] OR "West Sahara" [Title/Abstract] OR Mauritania*[Title/Abstract]) AND ("sharp injur*"[Title/Abstract] OR "sharps injur*"[Title/Abstract] OR "sharp instrument injur*"[Title/Abstract] OR "sharp instrument wound*"[Title/Abstract] OR "sharp object injur*"[Title/Abstract] OR "sharp objects injur*"[Title/Abstract] OR "sharp object wound*"[Title/Abstract] OR "sharp objects wound *"[Title/Abstract] OR "Needlestick Injuries"[Mesh] OR "needle stick*"[Title/Abstract] OR needlestick*[Title/Abstract] OR "needle-stick*"[Title/Abstract] OR "needle wound*"[Title/Abstract] OR "needle Injur*"[Title/Abstract] OR "Wounds, Stab"[Mesh:NoExp] OR "stab-wound*"[Title/Abstract] OR "stab-injur*"[Title/Abstract] OR "stab wound*"[Title/Abstract] OR "stab injur*"[Title/Abstract] OR "stabbing wound*"[Title/Abstract] OR "stabbing injur*"[Title/Abstract] OR "Wounds, Penetrating"[Mesh:NoExp] OR "penetrating wound*"[Title/Abstract] OR "penetrating injur*"[Title/Abstract] OR "penetration wound*"[Title/Abstract] OR "penetration injur*"[Title/Abstract] OR "body fluid exposure*"[Title/Abstract] OR "body fluids exposure*"[Title/Abstract] OR "bodily fluid exposure*"[Title/Abstract] OR "bodily fluid exposure*"[Title/Abstract] OR "biofluid exposure*"[Title/Abstract] OR "biofluids exposure*"[Title/Abstract] OR "splash exposure*"[Title/Abstract] OR "blood exposure*"[Title/Abstract] OR "body fluid splash*"[Title/Abstract] OR "blood splash*"[Title/Abstract])) AND ("Health Personnel"[Mesh] OR "health personnel*"[Title/Abstract] OR "hospital personnel*"[Title/Abstract] OR "health worker*"[Title/Abstract] OR "health care"[Title/Abstract] OR healthcare[Title/Abstract] OR "health-care"[Title/Abstract] OR "medical staff"[Title/Abstract] OR "health occupation*"[Title/Abstract] OR physician*[Title/Abstract] OR doctor*[Title/Abstract] OR nurse*[Title/Abstract] OR nursing[Title/Abstract] OR pharmac*[Title/Abstract] OR "physical therap*"[Title/Abstract] OR "occupational therap*"[Title/Abstract] OR dental*[Title/Abstract] OR dentist*[Title/Abstract] OR "Faculty"[Mesh] OR "medical facult*"[Title/Abstract] OR "health educator*"[Title/Abstract] OR "Students, Health Occupations"[Mesh] OR "medical student*"[Title/Abstract] OR "premedical student*"[Title/Abstract] OR "pre-medical student*"[Title/Abstract] OR "medical graduate*"[Title/Abstract] OR "public health student*"[Title/Abstract] OR "medical intern*"[Title/Abstract] OR "medical residen*"[Title/Abstract] OR anatomist*[Title/Abstract] OR anesthetist*[Title/Abstract] OR anesthesiolog*[Title/Abstract] OR endodontist*[Title/Abstract] OR orthodontist*[Title/Abstract] OR epidemiologist*[Title/Abstract] OR "infection control practitioner*"[Title/Abstract] OR nutritionist*[Title/Abstract] OR allergist*[Title/Abstract] OR cardiolog*[Title/Abstract] OR dermatolog*[Title/Abstract] OR endocrinolog*[Title/Abstract] OR gastroenterolog*[Title/Abstract] OR "general practitioner*"[Title/Abstract] OR geriatric*[Title/Abstract] OR nephrolog*[Title/Abstract] OR neurolog*[Title/Abstract] OR obstetric*[Title/Abstract] OR oncolog*[Title/Abstract] OR ophthalmolog*[Title/Abstract] OR otolaryngolog*[Title/Abstract] OR patholog*[Title/Abstract] OR

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pediatric*[Title/Abstract] OR paediatric*[Title/Abstract] OR neonatal[Title/Abstract] OR physiatrist*[Title/Abstract] OR pulmonolog*[Title/Abstract] OR radiolog*[Title/Abstract] OR rheumatology*[Title/Abstract] OR surgeon*[Title/Abstract] OR midwife*[Title/Abstract] OR urolog*[Title/Abstract] OR "medical technician*" OR "sterilisation technician*"[Title/Abstract] OR "sterilisation assistant*"[Title/Abstract] OR "sterilisation staff"[Title/Abstract] OR "sterilization technician*"[Title/Abstract] OR "sterilization assistant*"[Title/Abstract] OR "sterilization staff"[Title/Abstract] OR "sterilizing technician*"[Title/Abstract] OR "sterilizing assistant*"[Title/Abstract] OR "sterilizing staff"[Title/Abstract] OR "disinfection technician*"[Title/Abstract] OR "disinfection assistant*"[Title/Abstract] OR "disinfection staff"[Title/Abstract] OR "sterilisation personnel*"[Title/Abstract] OR "sterilization personnel*"[Title/Abstract] OR caregiver*[Title/Abstract] OR "Housekeeping, Hospital"[Mesh] OR "Hospitals"[Mesh] OR "Health Facilities"[Mesh:NoExp] OR "Laboratories"[Mesh] OR "Dental Facilities"[Mesh] OR hospital[Title/Abstract] OR hospitals[Title/Abstract] OR "medical center*"[Title/Abstract] OR "care center*"[Title/Abstract] OR "medical facility*"[Title/Abstract] OR "health facilit*"[Title/Abstract] OR "medical care facility*"[Title/Abstract] OR "medical care center*"[Title/Abstract] OR "health unit*"[Title/Abstract] OR "health center*"[Title/Abstract] OR "ambulatory care*"[Title/Abstract] OR clinic[Title/Abstract] OR clinics[Title/Abstract] OR laboratories[Title/Abstract] OR laboratory[Title/Abstract] OR "medical ward*"[Title/Abstract] OR "critical care unit"[Title/Abstract] OR "intensive care unit"[Title/Abstract] OR "ER"[Title/Abstract] OR "emergency center*"[Title/Abstract] OR "emergency department"[Title/Abstract] OR "emergency unit"[Title/Abstract] OR "delivery room*"[Title/Abstract] OR "sterilization room*"[Title/Abstract] OR "sterilisation room*"[Title/Abstract] OR "sterilizing room*"[Title/Abstract] OR "sterilising room*"[Title/Abstract] OR "disinfection room*"[Title/Abstract] OR "operation room*"[Title/Abstract] OR "operating theater*"[Title/Abstract] OR "trauma center*"[Title/Abstract])

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Sharps injuries and splash exposures among healthcare workers in Arab countries: protocol of a systematic review and meta-analysis

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Sharps injuries and splash exposures among healthcare workers in Arab countries: protocol of a systematic review and meta-analysis

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ABSTRACT

Introduction Sharps injuries, including needlestick injuries, and splash exposures constitute serious occupational health problems for healthcare workers, carrying the risk of blood-borne infections. However, data on such occupational incidents and their risk factors in healthcare settings are scarce and not systematically summarized in the Arab countries.

The aim of this study is to conduct a systematic review and meta-analysis to review published literature about sharps injuries and splash exposures of healthcare workers in Arab countries, with the objectives to determine the incidence and/or prevalence of these events, their identified risk factors and the applied preventive and postexposure prophylactic measures.

Methods and analysis The protocol is developed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocol (PRISMA-P) guidelines. A comprehensive pre-search developed in January to March 2021 in the database PubMed will be followed by a systematic search of six, core medical and health science databases; PubMed, EMBASE, Scopus, CINAHL, Web of Science and Africa-Wide Information in May 2021. The search will be performed without any filters or restrictions for publication years. Covidence systematic review tool will be used for document management, blinded screening and study selection. Two reviewers will independently screen the records, extract data and conduct risk of bias assessment. Results will be synthesized narratively in summary tables, and, if findings allow, meta-analysis will be conducted on the incidence and/or prevalence of sharps injuries and splash exposures, and on the effect size of risk factors.

Ethics and dissemination The systematic review methodology does not require ethics approval due to the nature of the study design based only on published studies. The results of the systematic review will be published in a peer-reviewed journal, disseminated to stakeholders and made publicly available.

Prospero registration number CRD42021242416

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Strengths and limitations of this study

- The study will rigorously follow the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
- The study will use transparent screening, data extraction and analysis methodology carried out by several researchers knowledgeable in conducting systematic reviews.
- Meta-analysis, including subgroup and sensitivity analysis, will be conducted if suitable results can allow for that.
- Scarcity of data and high clinical heterogeneity can be the main limitations of the study.

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INTRODUCTION

A sharps injury is a penetrating stab wound from a sharp object, such as a needlestick injury.[1] It is a very common event among healthcare personnel and, together with splash exposures on mucus membranes, constitute the most efficient method for transmitting blood-borne infectious agents from patients to healthcare staff. Sharps injuries are primarily associated with occupational transmission of hepatitis B (HBV), hepatitis C (HCV) and human immunodeficiency virus (HIV), but may also play role in the spread of several other infectious agents.[2] It is a feared, although still only hypothetical, route of transmission of coronaviruses, as well.[3] Health care personnel are at increased risk of sharps injuries and splash exposures because of the high frequency of contacting blood and other body fluids and potentially contaminated sharp objects.[4] As a consequence of the disease burden of occupational blood-borne infections, such incidents can entail a number of different costs including the loss of employee time, the cost of tying up staff to investigate the injury, expense of laboratory testing, the cost of postexposure prophylaxis and treatment for exposed and infected workers, expenses of temporarily or permanently replacing staff, and so on.[1,2,4,5]

According to the World Health Report 2002 by the World Health Organization,[6] about two million healthcare workers sustain percutaneous exposure to infectious diseases every year. The highest frequency of needlestick injuries has been reported among nurses, because they are the most exposed to needles and other sharp instruments during their work.[7]

There are several studies, mostly published on sharps injuries in healthcare settings from Europe and America. A comprehensive study estimated annual number of sharps injuries to be 384,000 in the United States, 100,000 in the United Kingdom, 700,000 in Germany, 29,719 in France, 28,200 in Italy and 21,815 in Spain.[68] There are reports about the high frequency of sharps injuries from other parts of the world, too. Nearly one out of five healthcare workers had experienced needlestick or sharps injury at least once within the past one year according to the results of a cross-sectional study that was conducted in 2014 in four hospitals in Southeast Ethiopia.[9] In a large-scale hospital-based survey in China, 6.3% of healthcare workers sustained sharps injury in the last month, corresponding to 1,032 incidents per 1,000 employees yearly.[10] The majority of injuries were caused by hollow-bore needles (63%) but only 4.3% of the incidents were reported to the infection control team. Another, yet unpublished study from China also found a high (41%) 5-year prevalence of sharps injuries among hospital personnel and a substantial (over 80%) underreporting was revealed (unpublished data).

However, limited information exists about sharps injuries and splash exposures of healthcare workers in the Middle East and North Africa. Several studies were published on this topic from Iran and their results were summarized in a systematic review.[11] However, the information on sharps injuries and splash exposures in Arab countries is limited and it has never been systematically reviewed. The results of an

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anonymized questionnaire survey among healthcare workers in the United Arab Emirates showed that 19% of the respondents had sustained a sharps injury in the calendar year 2006 and 53% of these sharps injuries were caused by blood contaminated sharp objects.[12] In a 2003 Egyptian study conducted among 1485 healthcare workers, 35.6% were found to be exposed to needlestick injury during the past three months, with an estimated annual number of nearly five needlesticks per worker.[13]

The Arab World consists of 22 countries in the Middle East and North Africa: Algeria, Bahrain, the Comoros Islands, Djibouti, Egypt, Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, Mauritania, Oman, Palestine, Qatar, Saudi Arabia, Somalia, Sudan, Syria, Tunisia, the United Arab Emirates, and Yemen. Although Arab countries constitute a heterogeneous group by economic and social development and consequently by the structure and development of their healthcare systems, it is a general phenomenon that healthcare employees work in fast-paced, stressful and frequently understaffed environment, that has also been affected by the ongoing COVID-19 pandemic. They have to perform various tasks, use sharp objects and can get easily into contact with body fluids. Their risk for sustaining sharps injury or splash exposure is high and probably severely underreported. To improve the occupational health and safety management system of this important workplace hazard, the systematic review and synthesis of the limited existing information from the region is indispensable.

OBJECTIVES

The aim of the study is to perform a systematic review by collecting and synthetizing the information available in the scientific literature about occupational sharps injuries, including needlestick injuries, and splash exposures occurring to healthcare workers engaged in clinical patient care in Arab countries, and to conduct meta-analysis if findings make it possible.

To achieve the aim, the following objectives will be pursued:

1. To conduct a systematic review of in extenso research articles published in peer-reviewed scientific journals in English or Arabic on occupational sharps injuries and splash exposures among healthcare workers engaged in clinical patient care in Arab countries.
2. To summarize incidence and/or prevalence of occupational sharps injuries and splash exposures among healthcare workers engaged in clinical patient care in Arab countries.
3. To identify risk factors statistically significantly associated with the occurrence of occupational sharps injuries and splash exposures.
4. To summarize the existing preventive and postexposure prophylactic measures used in the management of sharps injuries and splash exposures in healthcare settings in Arab countries.

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If findings allow for conducting meta-analysis, the following objectives will also be fulfilled:

5. To compute pooled frequency estimates for the incidence and/or prevalence of occupational sharps injuries and splash exposures among healthcare workers engaged in clinical patient care in Arab countries, with the possibility of subgroup analysis.
6. To compute pooled effect size estimates for potential risk factors of occupational sharps injuries and splash exposures, with the possibility of subgroup analysis.

METHODS AND ANALYSIS

This protocol follows the Preferred Reporting Items for Systematic Reviews and Meta-analyses Protocol (PRISMA-P) guidelines.[14] The completed PRISMA-P check list is available in Supplementary file 1. The protocol is registered online on the international prospective register of systematic reviews, PROSPERO: CRD42021242416. The planned start and end dates of the study are 1th of May 2021 and 30th of April 2022. The final review will be informed by Cochrane Handbook for Systematic Reviews of Interventions and follow the new PRISMA 2020 statement.[15 16]

Eligibility criteria

A PECO (population, exposure, comparator and outcomes) statement is developed to review information on occupational sharps injuries and splash exposures occurring to healthcare workers engaged in clinical patient care in Arab countries.

Population

Inclusion: All workers employed in any full- or part-time jobs, including students training, performing clinical care for patients, sterilization or cleaning activities in in- or outpatient healthcare settings (referred to as healthcare workers) located in Arab countries.

Exclusion: Workers not performing clinical care, sterilization or cleaning activities in in- or outpatient healthcare settings, and working in other than Arab countries.

Exposure

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Inclusion: Workplace accidents caused by sharp objects, including needles, or by splash of liquid potentially contaminated with body fluids, occurring in healthcare settings, and the risk factors of such accidents.

Exclusion: Accidents occurring not in healthcare settings, not caused by sharp objects or by splash of liquid potentially contaminated with body fluids.

Comparator

There will be no comparator for the main outcome of the review as it is the incidence/prevalence of occupational sharps injuries and splash exposures among healthcare workers in Arab countries. Also, there will be no comparator for the additional outcome of preventive and postexposure prophylactic measures used in the management of sharps injuries and splash exposures in healthcare settings in Arab countries.

The comparator for the additional outcome of risk factors of sharp injuries and splash exposures will be healthcare workers who have not sustained sharps injury or splash exposure.

Outcome

Injuries in the form of penetrating stab wounds on any sites of the body, and splash exposures, sustained by healthcare workers during an occupational accident caused by sharp objects, primarily needles, used in the work process or being present in the workplace, or by splash of liquids potentially contaminated with body fluids during the work process. The main outcome of the review will be the incidence and/or prevalence of occupational sharps, including needlestick, injuries and splash exposures among healthcare workers in Arab countries. Additional outcomes will be risk factors identified by statistical analysis to be significantly correlated to the occurrence of occupational sharps injuries and splash exposures in healthcare settings, and their effect size; and preventive and postexposure prophylactic measures used in the management of sharps injuries and splash exposures in healthcare settings in Arab countries.

Types of studies

Observational occupational epidemiological studies will be used that apply collection of individual data to determine the incidence and/or prevalence (cohort and cross-sectional studies) and identify risk factors (cohort and case-control studies) of occupational sharps injuries and splash exposures among healthcare workers in Arab countries. Other descriptive observational study designs (ecological, case series, case report) will also be included to review the preventive and postexposure prophylactic measures.

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Information sources

PubMed (NML), Scopus (Elsevier), Web of Science (Core Collection), EMBASE (Elsevier), CHNAHL (EBSCO) and Africa-Wide Information (EBSCO) electronic databases will be systematically searched to identify eligible studies published in peer-reviewed scientific journals.

Search strategy

A comprehensive, systematic search for literature will be conducted in May 2021 by a medical librarian specialized in systematic reviews (LÖ). Preliminary search term combinations have been identified and tested based on the PECO statement and inclusion and exclusion criteria outlined above. Pre-searches in PubMed was performed in January-March 2021 (LÖ). PubMed's MeSH was used to support a systematic inclusion of the search terms and their variations. Subject experts (BA, IK & IE) reviewed the final search term selection. The search string developed in PubMed will later be translated from English to Arabic and applied in all selected databases without geographical or publication year restrictions. We will use a combination of the search fields "Title/Abstract" and MeSH (alternatively Thesaurus or Subject Headings) for the best possible information retrieval. A search field coveting 'Title", "Abstract" and "Keywords" will be used in the absence of a MeSH, Thesaurus or Subject Headings.

The pre-search in PubMed can be found in Supplementary file 2. Reproducible search strings with result and notes for all databases included in the review will be appended to the final study. De-duplication and screening details will be reported in a PRISMA flow-diagram. A complete search-update of all databases will be performed before the final analysis and hand screening of the references lists in the included studies conducted.

Study records

Data management

The citations of the search results will be imported into the systematic review software Covidence for management and blinded screening of all records identified in the literature search.[17] The modules for automatic de-duplication, title/abstract and full text screening and blinded conflict resolving will support our work.

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Selection process

The screening and selection process will be documented in a PRISMA flow-diagram.[14] After automatic duplication removal in Covidence, the unique studies retrieved will be screened in two stages by two independent reviewers (IK and IE) based on the predetermined inclusion and exclusion criteria. In the first stage, title and abstract of publications will be screened. In the second stage, the full texts of publications selected during the first stage will be considered for inclusion also by two independent reviewers (IK and IE). Reason for exclusion will be documented in a PRISMA flow diagram. Discrepancies between the judgements of the two reviewers regarding the eligibility of studies and reason for exclusion will be resolved in the software by a third reviewer (BA). The screening and conflict resolving modules in Covidence is blinded.

Data collection process

Two reviewers (IK and IE) will independently extract data using a data extraction sheet developed in Excel 2019 for this study and pilot tested before use. Discrepancies will be resolved and extracted data validated through discussions with other reviewers (BA, MSP, RHA) until convergence and agreement is reached.

Data items

At a minimum, information will be extracted on publication characteristics (title, name of the first author, year of publication, DOI), study design (study type, risk factors, period of data collection, type of measurement of exposure and outcome, statistical analysis performed and adjustment to confounders), study population (demographic and occupational characteristics), results (incidence/prevalence of sharps injuries and splash exposures, crude and adjusted effect estimates of association identified as risk factors, preventive and postexposure prophylactic measures), and conflict of interest (declaration, funding).

Risk of bias in individual studies

The risk of bias in the included studies will be assessed using a modified version of the Navigation Guide RoB tool that is tailored specifically to systematic reviews of occupational health studies.[18] The Navigation Guide RoB tool assesses risk of bias in the following domains: selection bias, ascertainment bias, accuracy of exposure and outcome assessment, confounding, incomplete data and selective reporting, conflict of interest, and other biases, which will be completed with domains from the RoB-SPEO tool that was developed for the WHO/ILO Joint Estimates of the Work-related Burden of Disease and Injury project to

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assess risk of bias in studies estimating the prevalence of exposure to occupational risk factors.[19] Two reviewers (MSP and RHA) will independently assess the risk of bias in the included studies. Discrepancies between the judgements of the two reviewers regarding risk of bias of studies will be resolved by a third reviewer (BA).

Data synthesis

First, study findings will be synthesized narratively in summary tables regarding the main (incidence and prevalence of occupational sharps injuries and splash exposures) and the additional outcomes (risk factors and preventive and postexposure prophylactic measures).

If findings allow, meta-analysis will be conducted on the incidence and prevalence of occupational sharps injuries and splash exposures, and on potential risk factors. If we find two or more studies with eligible estimates on outcome frequency and/or on risk factor effect, two reviewers will independently investigate the clinical heterogeneity of the studies. If frequency and/or effect estimates are found clinically homogenous, then studies will be combined into one pooled frequency and/or effect estimate. Statistical heterogeneity of the studies will be tested using the I^2 statistic.[20] If two or more clinically homogenous studies are found to be sufficiently statistically homogenous to be combined in a meta-analysis, the frequency and/or effect estimates will be pooled in a quantitative meta-analysis, using the inverse variance method with a random-effects model to account for cross-study heterogeneity.[18] The meta-analysis will be conducted in RevMan software.[21] The pooled estimates will be presented in forest plots.

Analysis of subgroups or subsets

If there is evidence for differences in frequency and/or effect estimates by population (country, sex, age, job, healthcare sector, or by a combination of these variables), sub-group analyses by the relevant variable or combination of variables, will be conducted. Sensitivity analyses will be performed including studies judged to be of "low" or "probably low" risk of bias and judged to be of "low" or "probably low" risk of bias from conflict of interest.

Quality of cumulative evidence

A minimum of four reviewers will assess the quality of evidence for the entire body of evidence using the Navigation Guide quality of evidence assessment tool that is based on the GRADE approach and modified to be applicable in the fields of occupational and environmental health.[22-24]

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Patient and public involvement: No patient or members of the public will be included in this study.

Ethics and dissemination: Due to the nature of this study as a systematic review, ethics approval is not necessary as no patient or members of the public will be included in the study.

The results of this review will be published in a peer-reviewed journal and will be made publicly available in electronic format. In addition, if the findings of the review warrant a change in the practice of sharps injury and splash exposure management, a summary report will be disseminated to leading healthcare and occupational health policy makers in the East Mediterranean Region of the World Health Organization, and in the United Arab Emirates.

Contributors: BA, IK, IE, MSP, RHA, MSH and LÖ were involved in conceptualization and protocol development. BA, LÖ, IK and IE have developed the search strategy. Literature search will be conducted by LÖ and IK. Screening will be performed by IK and IE, with BA resolving conflicts. IK and IE will extract data, which will be validated by BA, MSP and RHA.

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Competing interests: None declared.

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Supplementary file 1

Reporting checklist for protocol of a systematic review and meta-analysis

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support			
Sources	#5a	Indicate sources of financial or other support for the review	N/A
Sponsor	#5b	Provide name for the review funder and / or sponsor	N/A
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	N/A
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	5
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	5,6

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Methods

Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6,7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	8
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9,10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10

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Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9,10
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10

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Supplementary file 2

Pre-search in PubMed

Source: PubMed (NLM)

Search date: 2021-03-11

Search specifications: All search terms are searched in the search fields: “title” and “abstract” (here marked with TI/AB) and in MeSH (when available). No filters or limitations applied.

Result: 309 records

Search string

(("Middle East"[Mesh:NoExp] OR "Bahrain"[Mesh] OR "Iraq"[Mesh] OR "Jordan"[Mesh] OR "Kuwait"[Mesh] OR "Lebanon"[Mesh] OR "Oman"[Mesh] OR "Qatar"[Mesh] OR "Saudi Arabia"[Mesh] OR "Syria"[Mesh] OR "United Arab Emirates"[Mesh] OR "Yemen"[Mesh] OR "Arab World"[Mesh] OR "Africa, Northern"[Mesh] OR "Somalia"[Mesh] OR "Djibouti"[Mesh] OR "Comoros"[Mesh] OR "Sudan"[Mesh] OR "Mauritania"[Mesh] OR "Middle East*"[Title/Abstract] OR "Near East"[Title/Abstract] OR "East Mediterranean" [Title/Abstract] OR "Eastern Mediterranean"[Title/Abstract] OR Arabic[Title/Abstract] OR Arabs[Title/Abstract] OR Arab[Title/Abstract] OR MENA[Title/Abstract] OR "Arabian Peninsula"[Title/Abstract] OR "North Africa" [Title/Abstract] OR "Northern Africa"[Title/Abstract] OR Maghreb[Title/Abstract] OR Maghrib[Title/Abstract] OR Egypt*[Title/Abstract] OR Jordan*[Title/Abstract] OR Lebanon[Title/Abstract] OR Lebanese[Title/Abstract] OR Syria*[Title/Abstract] OR Algeri*[Title/Abstract] OR Libya*[Title/Abstract] OR Morocco*[Title/Abstract] OR Tunis*[Title/Abstract] OR UAE[Title/Abstract] OR "United Arab Emirates"[Title/Abstract] OR Saudi*[Title/Abstract] OR KSA[Title/Abstract] OR Qatar*[Title/Abstract] OR Oman*[Title/Abstract] OR Emirat*[Title/Abstract] OR Yemen*[Title/Abstract] OR Kuwait*[Title/Abstract] OR Bahrain*[Title/Abstract] OR Somail*[Title/Abstract] OR Djibouti*[Title/Abstract] OR Comoro*[Title/Abstract] OR Comorian*[Title/Abstract] OR Iraq*[Title/Abstract] OR Palestin*[Title/Abstract] OR “West Bank” [Title/Abstract] OR “Gaza Strip” [Title/Abstract] OR

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Sudan*[Title/Abstract] OR "West Sahara" [Title/Abstract] OR Mauritania*[Title/Abstract]) AND ("sharp injur*"[Title/Abstract] OR "sharps injur*"[Title/Abstract] OR "sharp instrument injur*"[Title/Abstract] OR "sharp instrument wound*"[Title/Abstract] OR "sharp object injur*"[Title/Abstract] OR "sharp objects injur*"[Title/Abstract] OR "sharp object wound*"[Title/Abstract] OR "sharp objects wound *"[Title/Abstract] OR "Needlestick Injuries"[Mesh] OR "needle stick*"[Title/Abstract] OR needlestick*[Title/Abstract] OR "needle-stick*"[Title/Abstract] OR "needle wound*"[Title/Abstract] OR "needle Injur*"[Title/Abstract] OR "Wounds, Stab"[Mesh:NoExp] OR "stab-wound*"[Title/Abstract] OR "stab-injur*"[Title/Abstract] OR "stab wound*"[Title/Abstract] OR "stab injur*"[Title/Abstract] OR "stabbing wound*"[Title/Abstract] OR "stabbing injur*"[Title/Abstract] OR "Wounds, Penetrating"[Mesh:NoExp] OR "penetrating wound*"[Title/Abstract] OR "penetrating injur*"[Title/Abstract] OR "penetration wound*"[Title/Abstract] OR "penetration injur*"[Title/Abstract] OR "body fluid exposure*"[Title/Abstract] OR "body fluids exposure*"[Title/Abstract] OR "bodily fluid exposure*"[Title/Abstract] OR "bodily fluid exposure*"[Title/Abstract] OR "biofluid exposure*"[Title/Abstract] OR "biofluids exposure*"[Title/Abstract] OR "splash exposure*"[Title/Abstract] OR "blood exposure*"[Title/Abstract] OR "body fluid splash*"[Title/Abstract] OR "blood splash*"[Title/Abstract])) AND ("Health Personnel"[Mesh] OR "health personnel*"[Title/Abstract] OR "hospital personnel*"[Title/Abstract] OR "health worker*"[Title/Abstract] OR "health care"[Title/Abstract] OR healthcare[Title/Abstract] OR "health-care"[Title/Abstract] OR "medical staff"[Title/Abstract] OR "health occupation*"[Title/Abstract] OR physician*[Title/Abstract] OR doctor*[Title/Abstract] OR nurse*[Title/Abstract] OR nursing[Title/Abstract] OR pharmac*[Title/Abstract] OR "physical therap*"[Title/Abstract] OR "occupational therap*"[Title/Abstract] OR dental*[Title/Abstract] OR dentist*[Title/Abstract] OR "Faculty"[Mesh] OR "medical facult*"[Title/Abstract] OR "health educator*"[Title/Abstract] OR "Students, Health Occupations"[Mesh] OR "medical student*"[Title/Abstract] OR "premedical student*"[Title/Abstract] OR "pre-medical student*"[Title/Abstract] OR "medical graduate*"[Title/Abstract] OR "public health student*"[Title/Abstract] OR "medical intern*"[Title/Abstract] OR "medical residen*"[Title/Abstract] OR anatomist*[Title/Abstract] OR anesthetist*[Title/Abstract] OR anesthesiolog*[Title/Abstract] OR endodontist*[Title/Abstract] OR orthodontist*[Title/Abstract] OR epidemiologist*[Title/Abstract] OR "infection control practitioner*"[Title/Abstract] OR nutritionist*[Title/Abstract] OR allergist*[Title/Abstract] OR cardiolog*[Title/Abstract] OR dermatolog*[Title/Abstract] OR endocrinolog*[Title/Abstract] OR gastroenterolog*[Title/Abstract] OR "general practitioner*"[Title/Abstract] OR geriatric*[Title/Abstract] OR nephrolog*[Title/Abstract] OR neurolog*[Title/Abstract] OR obstetric*[Title/Abstract] OR oncolog*[Title/Abstract] OR ophthalmolog*[Title/Abstract] OR otolaryngolog*[Title/Abstract] OR patholog*[Title/Abstract] OR

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pediatric*[Title/Abstract] OR paediatric*[Title/Abstract] OR neonatal[Title/Abstract] OR physiatrist*[Title/Abstract] OR pulmonolog*[Title/Abstract] OR radiolog*[Title/Abstract] OR rheumatology*[Title/Abstract] OR surgeon*[Title/Abstract] OR midwife*[Title/Abstract] OR urolog*[Title/Abstract] OR "medical technician*" OR "sterilisation technician*"[Title/Abstract] OR "sterilisation assistant*"[Title/Abstract] OR "sterilisation staff"[Title/Abstract] OR "sterilization technician*"[Title/Abstract] OR "sterilization assistant*"[Title/Abstract] OR "sterilization staff"[Title/Abstract] OR "sterilizing technician*"[Title/Abstract] OR "sterilizing assistant*"[Title/Abstract] OR "sterilizing staff"[Title/Abstract] OR "disinfection technician*"[Title/Abstract] OR "disinfection assistant*"[Title/Abstract] OR "disinfection staff"[Title/Abstract] OR "sterilisation personnel*"[Title/Abstract] OR "sterilization personnel*"[Title/Abstract] OR caregiver*[Title/Abstract] OR "Housekeeping, Hospital"[Mesh] OR "Hospitals"[Mesh] OR "Health Facilities"[Mesh:NoExp] OR "Laboratories"[Mesh] OR "Dental Facilities"[Mesh] OR hospital[Title/Abstract] OR hospitals[Title/Abstract] OR "medical center*"[Title/Abstract] OR "care center*"[Title/Abstract] OR "medical facility*"[Title/Abstract] OR "health facilit*"[Title/Abstract] OR "medical care facility*"[Title/Abstract] OR "medical care center*"[Title/Abstract] OR "health unit*"[Title/Abstract] OR "health center*"[Title/Abstract] OR "ambulatory care*"[Title/Abstract] OR clinic[Title/Abstract] OR clinics[Title/Abstract] OR laboratories[Title/Abstract] OR laboratory[Title/Abstract] OR "medical ward*"[Title/Abstract] OR "critical care unit"[Title/Abstract] OR "intensive care unit"[Title/Abstract] OR "ER"[Title/Abstract] OR "emergency center*"[Title/Abstract] OR "emergency department"[Title/Abstract] OR "emergency unit"[Title/Abstract] OR "delivery room*"[Title/Abstract] OR "sterilization room*"[Title/Abstract] OR "sterilisation room*"[Title/Abstract] OR "sterilizing room*"[Title/Abstract] OR "sterilising room*"[Title/Abstract] OR "disinfection room*"[Title/Abstract] OR "operation room*"[Title/Abstract] OR "operating theater*"[Title/Abstract] OR "trauma center*"[Title/Abstract])

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